basis $\{ |m\rangle \mid m = j - \ell \,,\ \ell = 0, 1, \dots, n-1 \}$ $\qquad j \in \mathbb{C}$

$J_z |m\rangle = m |m\rangle$

$J_+ |m\rangle \propto |m+1\rangle$

$J_- |m\rangle \propto |m-1\rangle$

---

$$J^2 = J_x^2 + J_y^2 + J_z^2 = J_z^2 + \frac{1}{2}\left( J_+ J_- + J_- J_+ \right)$$

$= J_- J_+ + [J_+, J_-]$

$[J^2, J_x] = [J^2, J_y] = [J^2, J_z] = 0$

<u>Shur's lemma:</u> $\left.\begin{array}{l} J^2 : V \to V \quad \text{preserves } \underline{so}(3) \text{ action} \\ V \text{ irreducible rep} \end{array}\right\} \Rightarrow J^2$ multiple of identity

---

$[J_+, J_-] = 2 J_z \quad \Rightarrow \quad J^2 = J_z (J_z + 1) + J_- J_+$

$J^2 |j\rangle = \lambda |j\rangle \qquad \lambda \in \mathbb{C} \qquad (J^2 = \lambda \mathbb{1})$

$$J^2 |j\rangle = J_z (J_z + 1) |j\rangle + J_- J_+ |j\rangle$$

$= 0$

$$= j(j+1) |j\rangle$$

$$\boxed{\lambda = j(j+1)}$$

$T_r(AB) = T_r(BA)$

$[J_+, J_-] = 2 J_z \qquad T_r(J_z) = \frac{1}{2} T_r(J_+ J_- - J_- J_+)$

$$= \frac{1}{2} \left\{ T_r(J_+ J_-) - T_r(J_- J_+) \right\}$$

$$= 0$$

$\frac{1}{2} n(n-1)$

$$T_r(J_z) = \text{sum of eigenvalues} = \sum_m m = \sum_{\ell=0}^{n-1} j - \ell = nj - \sum_{\ell=1}^{n-1} \ell$$

$$= \frac{1}{2} n (2j - n + 1) = 0$$

$$\boxed{n = 2j + 1}$$

$j = 0, \frac{1}{2}, 1, \frac{3}{2}, 2, \frac{5}{2}, \dots$ (half-integer)

$j=0$ 1-dim rep (trivial)

$j=1/2$ 2-dim rep spin rep.

$j=1$ 3-dim rep

$n = \dim V$

$m = -j, \dots, j$

$-j, -j+1, -j+2, \dots, j$

<u>let</u> $J_+ |m\rangle = B_+(m) |m+1\rangle$ $\qquad J_- |m\rangle = B_-(m) |m-1\rangle$

$B_+(j) = 0$

$B_-(-j) = 0$

<u>change basis:</u> $|\tilde{m}\rangle = \alpha(m) |m\rangle$ $\qquad J_z |\tilde{m}\rangle = m |\tilde{m}\rangle$

$$J_+ |\tilde{m}\rangle = \alpha(m) J_+ |m\rangle = \alpha(m) B_+(m) |m+1\rangle$$

$$= \underbrace{\frac{\alpha(m)}{\alpha(m+1)} B_+(m)}_{C_+(m)} |\widetilde{m+1}\rangle$$

$\alpha(m)$ arbitrary

choose so that $C_\pm(m)$ real and non-negative and $C_+(m) = C_-(m+1)$

How?

$$J^2 = J_z(J_z+1) + J_- J_+$$

$$J^2 |m\rangle = j(j+1) |m\rangle$$

$$\shortparallel$$

$$(J_z(J_z+1) + J_- J_+) |m\rangle$$

$$= m(m+1) |m\rangle + B_+(m) J_- |m+1\rangle$$

$$= (m(m+1) + B_+(m) B_-(m+1)) |\tilde{m}\rangle$$

$$C_+(m) = \frac{\alpha(m)}{\alpha(m+1)} B_+(m) \quad \xleftrightarrow{=} \quad C_-(m+1) = \frac{\alpha(m+1)}{\alpha(m)} B_-(m+1)$$

$$C_-(m) = \frac{\alpha(m)}{\alpha(m-1)} B_-(m)$$

choose $\alpha(j) = 1$

$$\alpha(m)^2 = \alpha(m+1)^2 \frac{B_-(m+1)}{B_+(m)}$$

$$\alpha(j-1) = e^{i\theta(j-1)} \sqrt{\quad}$$

$\alpha(j-2)$ ---

---

choose inner product s.t. $\langle \tilde{m} | \tilde{m}' \rangle = \delta_{mm'}$

$J_z = J_z^\dagger$ and $J_+^\dagger = J_-$ $\qquad C_+(m) = C_-(m+1)$

$J_x^\dagger = J_x$

$J_y^\dagger = J_y$

$$J^2 |\tilde{m}\rangle = j(j+1) |m\rangle$$

$$\parallel$$

$$= \left( m(m+1) + C_+^2(m) \right) |\tilde{m}\rangle$$

$$C_+(m) = \sqrt{j(j+1) - m(m+1)}$$

$$C_-(m) = \sqrt{j(j+1) - m(m-1)}$$

---

labelled by $j = 0, 1/2, 1, 3/2$ --- $\qquad \dim = 2j+1$

eigenvalues of $J_z$ are $-j, -j+1, \ldots, j$

rep: action of $J_z, J_\pm$ on $V$

$J_z |\tilde{m}\rangle = m |\tilde{m}\rangle$

$J_\pm |\tilde{m}\rangle = C_\pm(m) |\tilde{m} \pm 1\rangle$

$|\tilde{m}\rangle \equiv |j\, m\rangle$

eigenvalue of $J_z$

label of rep

angular momentum op: $\vec{J} = \vec{x} \times \vec{p}$

$J_x, J_y, J_z$

$[J_x, J_y] = i J_z$

$J^2$ length squared $= j(j+1)\mathbb{1}$

rotation around z-axis by $\theta$ degrees

$\rightarrow R(\theta) = e^{-i\theta J_z}$

$J_z |j\, m\rangle = m |j\, m\rangle$

$R(\theta) |j\, m\rangle = e^{-i\theta m} |j\, m\rangle$

to be a rep, we need $R(2\pi)$ to act as identity

$R(2\pi) |j\, j\rangle = e^{-i 2\pi j} |j\, j\rangle$

if $j = 1/2$ $\quad e^{-i2\pi j} = -1$

we actually want proj. reps.

What makes a vector a vector?

$\rightarrow$ 3D thing transforms in the appropriate way under the action of $SO(3)$

being at element of a 3-dim rep of $SO(3)$

Scalars $\rightarrow$ trivial rep.

spinors (in-between) 2D and transform like the 2D rep of $SO(3)$

or like $SU(2)$

(double cover of $SO(3)$)

run 2 tensors: $j=1 \otimes j=1$

9-dim object

$$j' \otimes j'' = \bigoplus_{j=|j'-j''|}^{j'+j''} j$$